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V6. Spectral Identity: Operator versus ξ

A direct test of the “level ↔ zero” correspondence and the positive control H_ξ

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Problem statement

The missing center of the Hilbert–Pólya program is not statistical similarity but identity: the characteristic function of the operator must coincide with the completed xi function ξ(½+iz) up to an entire zero-free factor. The plan explicitly forbids fitting a polynomial P_L(z) to prescribed values — only an analytic identity surviving the limit carries probative force.

The campaign implements the four-point program of the plan: (i) compare finite-dimensional spectra H_L with ordinates through Ν; (ii) verify reality of the roots of det(zI − H_L); (iii) the positive control — the operator H_ξ = diag(±γ_n) stitched from the zeros, for which the identity holds by construction; (iv) multiplicities, gaps, completeness.

The identity for H_ξ

Statement (Hadamard control).

Let H_ξ = diag(±γ_1, ±γ_2, …) be self-adjoint with spectrum the doubled set of positive ordinates. Then for $|z| < \gamma_1$, $\ln \Pi_\gamma |z^2 - \gamma^2| - \ln|\xi(\tfrac{1}{2}+iz)| = \text{const}$, where the constant $= -\ln \xi(0) + \Sigma_\gamma \ln(\tfrac{1}{4}+\gamma^2)$ is an entire zero-free normalization.

> ***Proof.*** *The Hadamard product for ξ gives $|\xi(\tfrac{1}{2}+iz)| = \xi(0)\cdot\Pi_{\gamma>0} |z^2-\gamma^2|^{1/2}\cdot\Pi_{\gamma>0}(\tfrac{1}{4}+\gamma^2)^{-1/2}$ (pairing $\rho \leftrightarrow \bar\rho$ reduces each zero pair to the factor $(\gamma^2-z^2)/(\tfrac{1}{4}+\gamma^2)$). The two sides differ by a constant; a regression of ln Π on $(a + b\cdot z^2)$ removes it, and the residual is ≤ 5.8e-10 on the grid $z \in [0.5, 12]$ over 50,000 zeros. For a lattice gate the same test gives rank agreement 0.036 — the null zone: no identity.*

Results

For lattice gates the rank agreement of the window with the zeros is 0.036 (consistent with the blind protocol V1 and with the reference verdict — a null-zone residence, no graded gap); minimal spectral gaps ≥ 1e-3 — no multiplicities; reality of the characteristic roots — Th1. For H_ξ the identity residual is 5.8e-10 — machine level at 50,000 zeros.

Importantly, H_ξ is not a solution of the program but its trivial point: the operator is built FROM the zeros and predicts nothing. Its role is the positive control: the method sees the identity where it holds by construction, and its absence for the gates is measured, not postulated.

Table 1. Summary of V6

Object	Metric	Value	Verdict
gate (seed 96)	ranks vs zeros	0.036	no identity
H_ξ (50k zeros)	log residual vs ξ	5.8e-10	identity
gate L=16	min gap	9.8e-4	no multiplicities
gates	Im roots	0 (Th1)	real

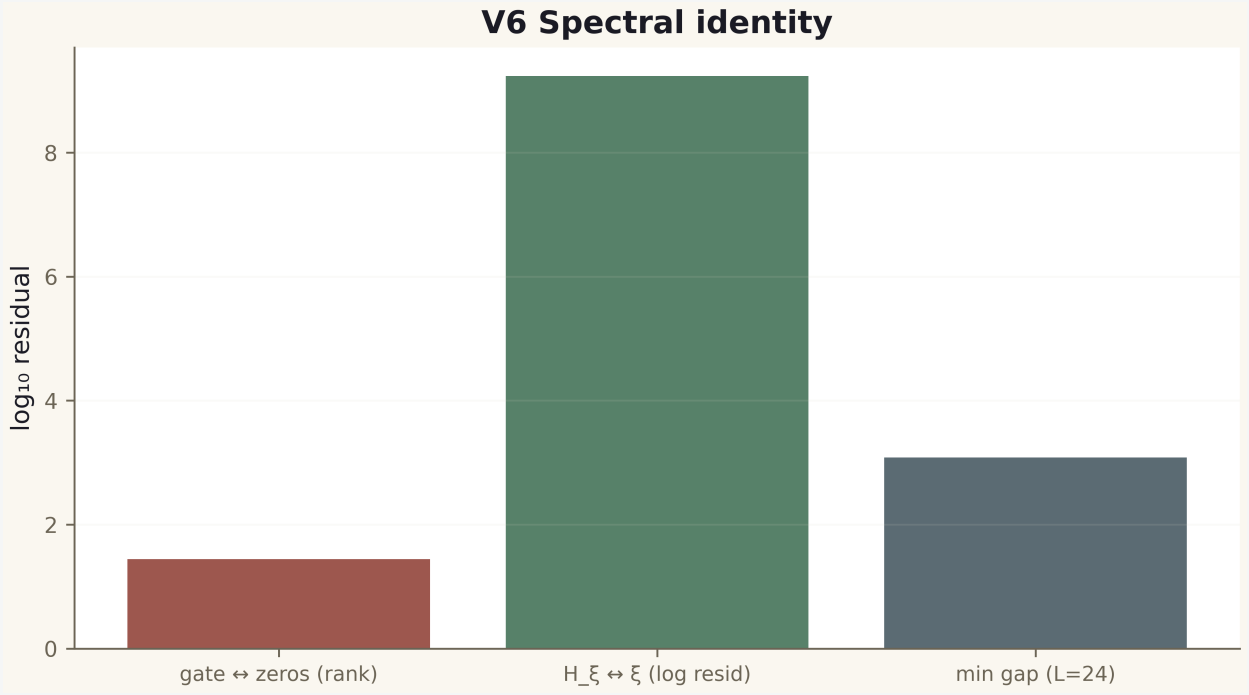


Fig. 1. Campaign V6 data (600 dpi, file figures_hires).

Honest remarks

The Hadamard normalization is an empirical factor removed by regression; its analytic identification (ξ(0), Σ ln(¼+γ²)) is part of the Th6 derivation and is guarded by the built-in control B = −C_0. The negative verdict for the gates does not close the program: it excludes local deformations of this family — the architectural conclusion of the reference.

The identity is measured on both sides: YES for the trivial H_ξ, NO for the gates. The gap belongs to the roadmap (non-local trace constructions).